

3. Confidence Scoring Architecture

In the Kalam Welfare Engine, absolute determinism is paramount. The system evaluates schemes strictly based on the extracted rules without LLM intervention.

Mathematical Formula

Confidence Score = (Number of Passed Rules) / (Total Number of Rules)

Core Logic Flow

- If ANY rule evaluates to explicitly FALSE (e.g. user age 28, required age ≤ 25), the scheme Confidence Score defaults to 0.0% and the verdict is absolute INELIGIBLE.
- If ALL rules evaluate to TRUE, the Confidence Score is 100.0% and verdict is ELIGIBLE.
- If rules are UNKNOWN or AMBIGUOUS, the score reflects the ratio of known passing rules, yielding a PARTIAL verdict.

Gaps Analysis

Whenever a partial score is returned, the engine attaches absolute "Gaps" identifying the specific boolean constraints blocking 100% eligibility.